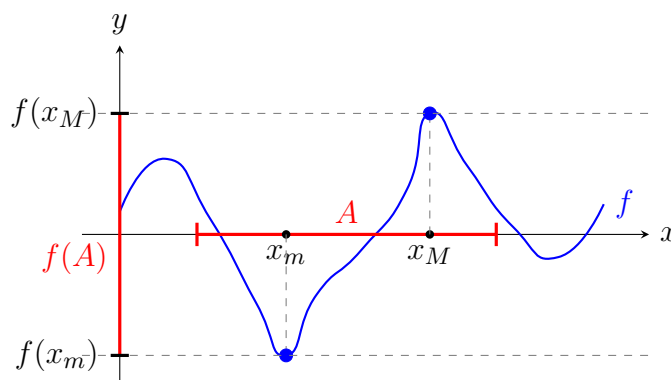




Maximum and minimum of a real-valued function

If $\sup_{x \in A} f(x) < +\infty$ and it belongs to $f(A)$ then it is the **maximum of f on A** . It is denoted

$$\max_A f \quad \text{or} \quad \max_{x \in A} f(x).$$

If $\inf_{x \in A} f(x) > -\infty$ and it belongs to $f(A)$ then it is the **minimum of f on A** . It is denoted

$$\min_A f \quad \text{or} \quad \min_{x \in A} f(x).$$

Since the minimum and the maximum of f on A belong to $f(A)$, there exist $x_m \in A$ and $x_M \in A$ such that

$$f(x_M) = \max_A f \quad \text{and} \quad f(x_m) = \min_A f.$$

Example 4.3: 1. For $\sin x$, we have

$$\max_{x \in \mathbb{R}} (\sin x) = 1 \quad \text{and} \quad \min_{x \in \mathbb{R}} (\sin x) = -1.$$

2. For x^2 we have

$$\sup_{x \in \mathbb{R}} (x^2) = +\infty \quad \text{and} \quad \min_{x \in \mathbb{R}} (x^2) = 0.$$

If $A = [-10, -3]$ then

$$\max_{x \in A} (x^2) = 100 \quad \text{and} \quad \inf_{x \in A} (x^2) = 9.$$

Note that in this last case, the infimum is not achieved, so there's no minimum on A .

4.3 Surjectivity, injectivity, and invertibility

Let us define some important *global* properties of functions, i.e. properties that tell us something about the function as a whole.

Definitions

Let $f : X \rightarrow Y$.

- We say that f is **surjective** (or **onto**) if $\text{im}(f) = Y$.
- We say that f is **injective** (or **1 – 1**, **one-to-one**) if for every $y \in Y$, the subset $f^{-1}(y) \subseteq X$ contains *at most* one element.
- If f is both surjective and injective, it is called a **bijection** (or a **bijjective function**).

Let us try to understand these concepts. Surjectivity means that for every $y \in Y$ there exists (at least) one $x \in X$ such that $f(x) = y$. See Figure 4.1.

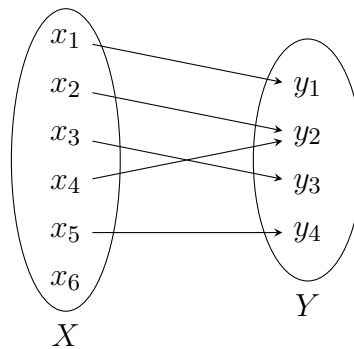


Figure 4.1: A surjective (*onto*) function

Injectivity means that every $y \in Y$ has at most one pre-image. So for any $y \in Y$, either there is no $x \in X$ such that $f(x) = y$ or there is one (and no more) $x \in X$ such that $f(x) = y$. Equivalently, if $x_1 \neq x_2$ then $f(x_1) \neq f(x_2)$. See Figure 4.2.

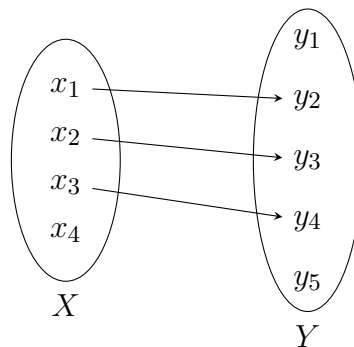


Figure 4.2: An injective (*1-1*) function

Finally, a bijective function f associates to every $x \in X$ exactly one $y \in Y$, and vice versa. See Figure 4.3. In this case we say that the sets X and Y are in **one-to-one (1-1) correspondence** under f .

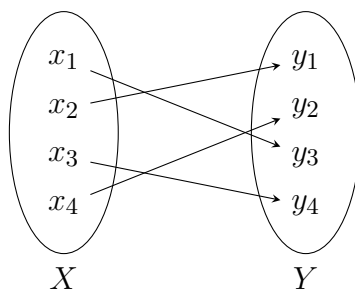


Figure 4.3: A bijective function

The case of a function $f : \mathbb{R} \rightarrow \mathbb{R}$

When $f : \mathbb{R} \rightarrow \mathbb{R}$ the aforementioned properties can be seen by looking at its graph in $\mathbb{R}^2$:

- f is *surjective* if its graph intersects any horizontal line at least once;
- f is *injective* if its graph intersects any horizontal line at most once;
- f is *bijective* if its graph intersects every horizontal line exactly once.

We observe that the arrows of an injective (1-1) function can be reversed to obtain a function from Y to X . So we define:

Inverse function

Let $f : X \rightarrow Y$ be one-to-one. We define its **inverse** $f^{-1} : \text{im}(f) \subseteq Y \rightarrow X$ as follows:

$$f^{-1}(y) = x \quad \text{where } x \text{ is the unique element in } X \text{ satisfying } f(x) = y.$$

Therefore any one-to-one function is also **invertible**. Observe that:

$$\text{dom}(f) = \text{im}(f^{-1}) \quad \text{and} \quad \text{im}(f) = \text{dom}(f^{-1})$$

Note that there is some abuse of notation¹ here: $f^{-1}(y)$ is defined to be the *element* $x \in X$, while before (see Section 4.2) we defined $f^{-1}(y)$ to be the *subset* $\{x\} \subseteq X$. In the case of a one-to-one function, this can be forgiven, because there is little practical difference between x and $\{x\}$.

The graph of the inverse function

Recall that the graph of $f : \text{dom}(f) \subseteq X \rightarrow Y$ is defined as

$$\Gamma(f) = \{(x, f(x)) \in X \times Y \mid x \in \text{dom}(f)\}.$$

¹“Abuse of notation” means that our notation is not entirely consistent.

This immediately implies that

$$\begin{aligned}\Gamma(f^{-1}) &= \{(y, f^{-1}(y)) \in Y \times X \mid y \in \text{dom}(f^{-1})\} \\ &= \{(f(x), x) \in Y \times X \mid x \in \text{dom}(f)\}.\end{aligned}$$

Comparing the two expressions for $\Gamma(f)$ and for $\Gamma(f^{-1})$, one can see that the graph of f^{-1} is obtained by mirroring the graph of f along the $x = y$ line. We can clearly see this for $f(x) = x^2$, $x \geq 0$, and its inverse $f^{-1}(x) = \sqrt{x}$, see Figure 4.4.

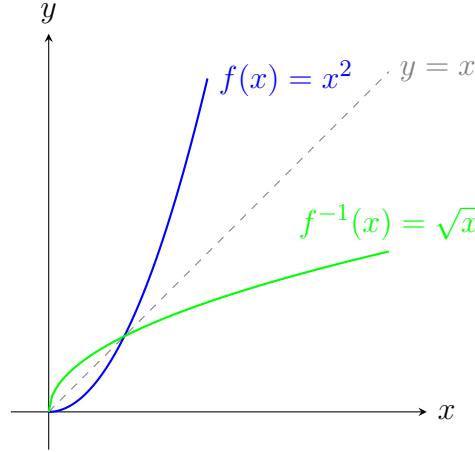


Figure 4.4: $f(x) = x^2$ and its inverse $f^{-1}(x) = \sqrt{x}$, defined on $x \geq 0$, are mirror images with respect to the line $x = y$

Sometimes we want to only look at part of the domain of a function. For example, in the example above, we looked at $f(x) = x^2$ only for $x \geq 0$, so that we could look at its inverse. Otherwise, if we had looked at $x \in \mathbb{R}$, then the preimage of any $y \geq 0$ is $\{+\sqrt{y}, -\sqrt{y}\}$ – i.e., there is no inverse function. What we did was to *restrict* $f(x) = x^2$ to $x \geq 0$:

Restriction of a function

Let $f : X \rightarrow Y$ be a function. Let $A \subseteq \text{dom}(f)$ be a subset of the domain of f . The restriction of f to A is a ‘new’ function $f|_A$ that is defined only on A , where it is identical to f :

$$f|_A : A \rightarrow Y \quad \text{defined as} \quad f|_A(x) = f(x), \quad \forall x \in A.$$

In Figure 4.4, the blue graph is the graph of the restriction of x^2 to $A = \{x \in \mathbb{R} \mid x \geq 0\}$.

4.4 Monotone functions

Functions that always increase/decrease are of particular interest because they might have important applications. For example, when you study *thermodynamics* you will see that **entropy** is a monotone increasing function of time, meaning that our world always becomes more disorganized.

Monotone functions $f : \mathbb{R} \rightarrow \mathbb{R}$

Monotone functions $f : \mathbb{R} \rightarrow \mathbb{R}$

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is said to be **(monotonically) increasing on** $I \subseteq \text{dom}(f)$ if for every $x_1, x_2 \in I$ with $x_1 < x_2$ we have $f(x_1) \leq f(x_2)$:

$$\forall x_1, x_2 \in I, \quad x_1 < x_2 \quad \Rightarrow \quad f(x_1) \leq f(x_2).$$

The function f is said to be **strictly increasing on** I if

$$\forall x_1, x_2 \in I, \quad x_1 < x_2 \quad \Rightarrow \quad f(x_1) < f(x_2).$$

Similarly, f is said to be **(monotonically) decreasing on** I if for every $x_1, x_2 \in I$ with $x_1 < x_2$ we have $f(x_1) \geq f(x_2)$:

$$\forall x_1, x_2 \in I, \quad x_1 < x_2 \quad \Rightarrow \quad f(x_1) \geq f(x_2).$$

The function f is said to be **strictly decreasing on** I if

$$\forall x_1, x_2 \in I, \quad x_1 < x_2 \quad \Rightarrow \quad f(x_1) > f(x_2).$$

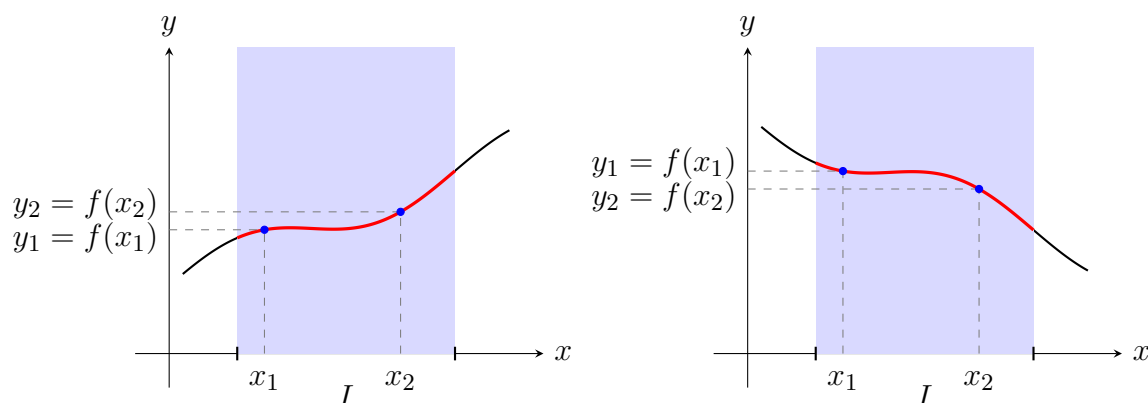


Figure 4.5: An increasing function (left) and a decreasing function (right)

Note that monotonically increasing/decreasing functions are allowed to have slope 0 and even to remain constant. So a constant function $f(x) = c$ is (trivially) both monotone increasing and monotone decreasing. Step functions such as $f(x) = \lceil x \rceil$ or $f(x) = \lfloor x \rfloor$ are monotonically increasing (but not decreasing).

Example 4.4: 1. $f(x) = c$ is monotonically increasing and decreasing.

2. $f(x) = x^2$ is neither increasing nor decreasing on $\mathbb{R}$.

3. $f(x) = x^2$ is strictly increasing on $[0, +\infty)$.

4. $f(x) = x^2$ is strictly decreasing on $(-\infty, 0]$.

5. $f(x) = x^3$ is strictly increasing on $\mathbb{R}$.

6. $f(x) = \begin{cases} 0 & \text{for } x < 0 \\ x^2 & \text{for } x \geq 0 \end{cases}$ is monotonically increasing on $\mathbb{R}$, and it is *strictly* increasing on $[0, +\infty)$.

Proposition 4.1: A function that is strictly increasing/decreasing on its domain is injective (one-to-one).

Proof. Consider first the case that $f : \text{dom}(f) \subseteq \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing on $\text{dom}(f)$. Let $x_1, x_2 \in \text{dom}(f)$ with $x_1 \neq x_2$. Without loss of generality $x_1 < x_2$, so that $f(x_1) < f(x_2)$. In particular, $f(x_1) \neq f(x_2)$, so that f is injective. The case of a strictly decreasing function follows the same idea of proof. $\square$

★★★ The converse statement – i.e. that an injective function is strictly monotone – is not true, see Figure 4.6 for a counterexample.

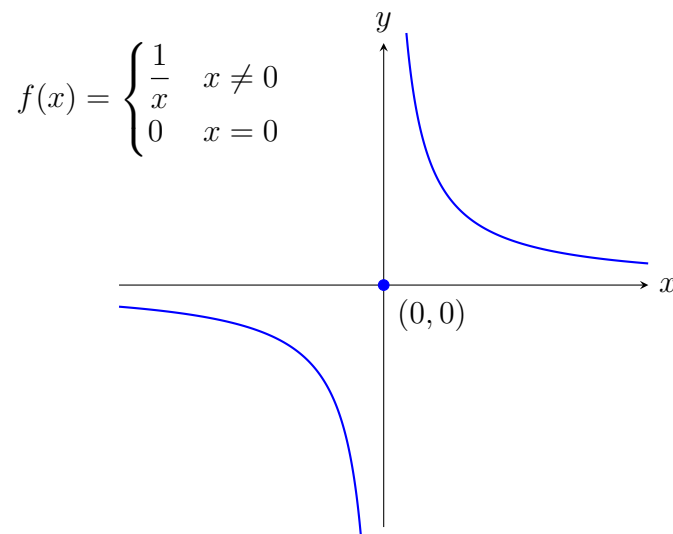


Figure 4.6: A one-to-one function that is neither increasing nor decreasing

Lemma 4.2: Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be monotonically increasing on some $A \subseteq \mathbb{R}$. Then $f + g$ is also monotonically increasing on A . If either f or g are *strictly* increasing on A , then so is $f + g$. The same statements hold if we replace everywhere the word ‘increasing’ with the word ‘decreasing’.

Proof. Exercise. $\square$

4.5 Composition of functions

The composition of functions – i.e. the application of two (or more) functions successively – is something that often comes up in mathematics and its applications.

Example 4.5 (Taxi fare): Suppose that the fare for riding a taxi is made of a flat fee of 3 Euros plus twice the distance travelled (in kilometers). So the fee for riding x kilometers is:

$$f(x) = 2x + 3.$$